Zero-Knowledge Across the Air-Gap: Verifiable Computation and Subliminal-Free Proving with Strictly Offline Private Witnesses

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Abstract

Zero-Knowledge Proof (ZKP) systems enable a prover to convince a verifier that a statement $\boldsymbol{x}$ is true with respect to a secret witness $\boldsymbol{w}$ without revealing any information about $\boldsymbol{w}$ across the verification channel. However, prevailing ZKP deployments rely on a flawed systemic assumption: that the proving environment itself is free from operating-system compromise or network-level exfiltration. Even in architectures utilizing hardware security modules or cold cryptographic wallets, "split-proving" protocols offload the computationally intensive witness generation and polynomial arithmetic onto a network-connected host, exposing the entirety of $\boldsymbol{w}$ in volatile memory (RAM). Conversely, strictly air-gapped systems—deployed across critical infrastructure, nuclear telemetry, and genomic data cleanrooms—achieve physical confidentiality of $\boldsymbol{w}$ at the cost of external auditability. In this paper, we introduce the formal framework for **Air-Gapped Zero-Knowledge Proving (AG-ZKP)**. We resolve the *Air-Gap Auditability Paradox* by synthesizing physical network isolation with succinct cryptographic verification over constrained, asymmetric physical channels (specifically optical QR fountain streams and unidirectional optical data diodes). Furthermore, we identify and mitigate three fundamental vulnerabilities unique to air-gapped proving: (1) *The Blinded Witness Synchronization Problem*, (2) *The Optical Bandwidth vs. Proof Complexity Bottleneck*, and (3) *Kleptographic Subliminal Channels* embedded within non-deterministic ZK proofs.

1. Introduction

1.1 The Networked Prover Vulnerability

Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be an NP relation defined by an arithmetic circuit $C : \mathbb{F}_p^n \times \mathbb{F}_p^m \to \mathbb{F}_p^l$. While the cryptographic zero-knowledge property guarantees that a generated proof $\pi \leftarrow \mathbf{Prove}(pk, x, w)$ leaks nothing about the private witness $w \in \mathcal{W}$ to a computationally bounded verifier $\mathcal{V}$, it offers zero protection during the execution of $\mathbf{Prove}(\cdot)$ itself.

In modern proving pipelines—spanning zk-SNARKs (e.g., Groth16, PLONK) and zk-STARKs (e.g., FRI-based polynomial protocols)—computing π requires expanding the compact secret witness w into a massive execution trace matrix $T \in \mathbb{F}_p^{N \times d}$ (where $N = 2^k \gg |w|$) and executing Number Theoretic Transforms (NTT/FFT) and Multi-Scalar Multiplications (MSM) across hundreds of megabytes or gigabytes of volatile system memory. Because resource-constrained secure elements (smartcards, cold hardware wallets) typically possess less than $\mathbf{1\,MB}$ of SRAM, existing protocols employ *delegated* or *split proving*: the isolated device retains only a 32-byte authorizing key $k_{\mathbf{auth}}$, while exporting the full private witness w (financial history, Merkle paths, biometric features, or

proprietary parameters) in plaintext to a network-connected host $\mathcal{H}_{\text{net}}$. Any kernel-level malware, spyware, or Direct Memory Access (DMA) adversary on $\mathcal{H}_{\text{net}}$ trivially extracts w from host RAM before π is ever constructed.

1.2 The Air-Gap Auditability Paradox

In high-assurance domains—such as industrial SCADA/nuclear facilities, defense intelligence enclaves, medical genomic repositories, and offline Root Certificate Authorities (CAs)—sensitive datasets w are stored exclusively inside physically air-gapped machines $\mathcal{M}_{\text{AG}}$ lacking RF transceivers or bidirectional network interfaces. This creates a fundamental conflict in systems security, which we term the **Air-Gap Auditability Paradox**:

- **Strict Physical Isolation** guarantees that w cannot be exfiltrated over a network, maximizing *Confidentiality*.
- **External Auditability** requires demonstrating to an external regulator, counterparty, or public ledger that the internal state w satisfies a compliance predicate $C(x, w) = 0$.

Historically, auditing an air-gapped system required either breaking the air-gap (introducing removable media), exposing raw subsets of w to external human auditors, or relying on physical procedural ceremonies (e.g., video-recorded key ceremonies). **Air-Gapped Zero-Knowledge Proving (AG-ZKP)** bridges this divide: w remains permanently confined within $\mathcal{M}_{\text{AG}}$, while only a succinct, zero-knowledge proof π traverses a constrained physical boundary.

2. Threat Model and System Definitions

We formalize the AG-ZKP architecture as an interaction across three distinct security domains:

1. **The Air-Gapped Prover ($\mathcal{P}_{\text{AG}}$):** A computationally capable offline computing node (e.g., an air-gapped ARM64/x86 appliance or mobile vault with $\geq 4\,\text{GB}$ volatile memory and physically removed RF/network interfaces) holding the long-term private witness w_{static}.
2. **The Untrusted Networked Bridge ($\mathcal{B}_{\text{net}}$):** An internet-connected relay terminal capable of communicating with $\mathcal{P}_{\text{AG}}$ solely through a constrained physical channel $\mathcal{C}_{\text{phys}}$ (specifically, a camera-to-screen optical link or a hardware-enforced unidirectional fiber-optic data diode).
3. **The External Verifier ($\mathcal{V}$):** A remote auditor or smart contract that receives (x, π) from $\mathcal{B}_{\text{net}}$ and outputs $\text{Verify}(vk, x, \pi) \in \{0, 1\}$.

Adversary Capabilities ($\mathcal{A}$): We assume a Byzantinely compromised bridge $\mathcal{B}_{\text{net}}$. The adversary $\mathcal{A}$ has full root control over $\mathcal{B}_{\text{net}}$, observes all inbound and outbound frames across $\mathcal{C}_{\text{phys}}$, and may attempt to feed malformed public state updates to $\mathcal{P}_{\text{AG}}$ or exploit covert channels within the proof π to infer w.

3. Overcoming the Core Challenges of Air-Gapped Proving

3.1 The Blinded Witness Synchronization Problem

In stateful ZKP applications, the full witness $w = (w_{\text{local}}, w_{\text{aux}})$ requires dynamic auxiliary data w_{aux} from the external world—for instance, a Merkle authentication path $\mathbf{Path}(l_i)$ proving that $\mathcal{P}_{\text{AG}}$'s secret record l_i resides in a global state root R_t. If $\mathcal{P}_{\text{AG}}$ requests $\mathbf{Path}(l_i)$ specifically from the compromised bridge $\mathcal{B}_{\text{net}}$, $\mathcal{B}_{\text{net}}$ immediately learns the leaf index i, violating metadata privacy across the ingress channel. We propose two mitigations:

- **Subtree k-Anonymity Optical Ingestion:** Rather than requesting a single branch of depth D, $\mathcal{B}_{\text{net}}$ transmits the truncated root path down to depth $D - d$, along with the complete dense subtree of size 2^d covering an anonymity set of $k = 2^d$ leaves via fountain-encoded optical frames.
- **Incremental Accumulator Folding:** Alternatively, $\mathcal{B}_{\text{net}}$ broadcasts universal state transitions $\Delta\mathbf{Acc}_t$, enabling $\mathcal{P}_{\text{AG}}$ to update its witness locally in $O(1)$ space without issuing any index-identifying query across the air-gap.

3.2 Bandwidth-Proving Analytical Optimization

Unlike gigabit network interfaces, physical air-gap channels exhibit severe bandwidth constraints. Let R_{in} denote the physical ingress rate (bytes/sec) and R_{out} denote the physical egress rate using Luby Transform (LT) or RaptorQ fountain codes (e.g., animated QR streams where $R_{\text{out}} \approx 2.0 - 3.5\ \text{KB/s}$). The total end-to-end latency T_{total} of an Air-Gapped ZK proof is governed by:

$$T_{\text{total}} = \frac{|x| + |w_{\text{aux}}|}{R_{\text{in}}} + T_{\text{prove}}(|C|, \mathbf{Arch}) + \frac{|\pi| \cdot (1 + \epsilon_{\text{FEC}})}{R_{\text{out}}}$$

where $|C|$ is the constraint count, $\mathbf{Arch}$ represents the offline compute capability, and $\epsilon_{\text{FEC}} \geq 0.15$ is the forward error correction overhead of the fountain stream. This latency model exposes a critical regime shift across physical channels:

Proof System	Proof Size $\lvert\pi\rvert$	Prover Time T_{prove} (2^{20} gates)	Optical QR Egress ($R_{\text{out}} = 2.5\ \text{KB/s}$)	Data Diode Egress ($R_{\text{out}} = 50\ \text{MB/s}$)	Optimal Air-Gap Domain
Groth16	$\approx 192\ \text{B}$	$\sim 4.5\ \text{s}$	$< 0.1\ \text{s}$ (1 Static QR)	$< 1\ \text{ms}$	Bidirectional Optical / Portable Air-Gap
PLONK (KZG)	$\approx 1.2\ \text{KB}$	$\sim 12.0\ \text{s}$	$\sim 0.6\ \text{s}$ (3–5 QR Frames)	$< 1\ \text{ms}$	Universal Circuit Optical Air-Gap
zk-STARK (FRI)	$\approx 85\ \text{KB}$	$\sim 2.1\ \text{s}$	$\sim 39.1\ \text{s}$ (~ 300 Frames)	$\sim 1.7\ \text{ms}$	Unidirectional Fiber Data Diodes & Post-Quantum Enclaves

Table 1: End-to-end latency trade-offs across optical QR fountain streams versus unidirectional fiber data diodes.

Key Finding: While FRI-based STARK provers execute faster in raw CPU time (T_{prove}), their larger proof size $|\pi|$ causes T_{total} on optical QR channels to be dominated by physical transmission latency. Conversely, over unidirectional fiber-optic data diodes (where R_{out} is high), STARKs dominate due to zero trusted setup and post-quantum resilience. Hybrid recursion (computing a STARK inside $\mathcal{P}_{\text{AG}}$ and wrapping its verification into a succinct Groth16/PLONK proof prior to optical egress) achieves the global latency minimum on optical links.

4. Subliminal-Free Proofs and Anti-Kleptography

A critical, previously unaddressed threat in air-gapped proving is **ZK Kleptography**. Standard zero-knowledge proving algorithms are probabilistic: the prover samples random blinding scalars $r = (r_1, \ldots, r_k) \xleftarrow{\$} \mathbb{F}_p^k$ to mask polynomial commitments. If the proving software or circuit binary loaded onto $\mathcal{P}_{\text{AG}}$ contains a supply-chain backdoor planted by $\mathcal{A}$, the malicious prover $\tilde{\mathcal{P}}_{\text{AG}}$ does not need a network connection to exfiltrate w. Instead, it computes $r_i = \text{PRF}_{k_{\text{adv}}}(w)$ such that the resulting proof π is mathematically valid ($\text{Verify}(vk, x, \pi) = 1$) and computationally indistinguishable from a random proof to any third party, yet the compromised bridge $\mathcal{B}_{\text{net}}$ holding k_{adv} extracts bits of the secret witness w directly from the group elements of π.

4.1 Deterministic Witness-Bound Derandomization

To eliminate kleptographic subliminal channels across the air-gap, we enforce **Strict Deterministic Proving**:

1. All blinding scalars r inside $\mathcal{P}_{\text{AG}}$ must be derived deterministically via a cryptographic sponge function (e.g., Poseidon2 or SHAKE-256) bound to a canonical commitment of the witness and a verifiable hardware seed s_{hw}:

$$r_i = \text{Hash}(\text{domain_sep} \parallel \text{Commit}(w) \parallel x \parallel i)$$

2. For proof systems supporting **Proof Rerandomization**, $\mathcal{P}_{\text{AG}}$ emits a canonical deterministic proof π_{det} (auditable against reproducible test vectors), and the untrusted bridge $\mathcal{B}_{\text{net}}$ applies public rerandomization $\pi' \leftarrow \text{Rerand}(pk, x, \pi_{\text{det}})$ outside the air-gap prior to external broadcast.

4.2 Zero-Remanence Witness Memory Scrubbing

During NTT/FFT polynomial interpolation, elements of w are algebraically diffused across $O(N)$ memory pages ("witness scattering"). An air-gapped prover must allocate the entire witness and execution trace buffer within a contiguous, page-locked virtual memory arena (`mlock` / non-swappable DRAM) and execute a cryptographic memory zeroization barrier (`explicit_bzero` followed by cache-line flush) prior to rendering the optical output frame.

5. High-Assurance Applications Beyond Finance

- **Unidirectional Data Diode Auditing (Nuclear & SCADA Telemetry):** Power grids and nuclear enrichment facilities stream sensor logs into an internal $\mathcal{P}_{\text{AG}}$. Every epoch, $\mathcal{P}_{\text{AG}}$ generates a STARK proof π_t asserting that operational parameters (temperature, centrifuge

velocity, radiation thresholds) remained strictly within treaty bounds $[L_{\min}, L_{\max}]$, transmitting π_t across a physical fiber-optic data diode to international inspectors without revealing plant schematics or raw telemetry.

- **Air-Gapped Genomic & Clinical Cleanrooms:** Biobanks hold raw human DNA sequences on offline servers. Researchers submit statistical query circuits via optical ingress; the server emits an AG-ZKP proving cohort correlations over authentic, signed genomic commitments without a single nucleotide sequence ever touching a networked machine.
- **Cryptographic Ceremonies & Offline Root CAs:** Replaces manual video-recorded human auditing in offline Hardware Security Module (HSM) ceremonies with continuous, zero-knowledge policy compliance proofs.

6. Conclusion

Physical air-gaps and Zero-Knowledge Proofs have historically evolved as separate disciplines—one physical, one mathematical. By uniting them under the **AG-ZKP** framework, we eliminate the single point of failure inherent in networked ZK proving (plaintext witness exposure in host RAM) while granting mathematically rigorous auditability to the world's most isolated computing environments.

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